

If statement:

```
a = 33
```

```
b = 200
```

```
if b > a:
```

```
    print("b is greater than a")
```

OUTPUT:

b is greater than a

else:

```
a = 200
```

```
b = 20
```

```
if b > a:
```

```
    print("b is greater than a")
```

```
elif a == b:
```

```
    print("a and b are equal")
```

```
else:
```

```
    print("a is greater than b")
```

OUTPUT:

a is greater than b

Python nested IF statements:

```
var = 100

if var < 200:

    print ("Expression value is less than 200")

    if var == 150:

        print ("Which is 150")

    elif var == 100:

        print ("Which is 100")

    elif var == 50:

        print ("Which is 50")

    elif var < 50:

        print ("Expression value is less than 50")

else:

    print ("Could not find true expression")


print ("Good bye!")
```

OUTPUT:

Expression value is less than 200

Which is 100

Good bye!

Loops:

while Loop Statements:

```
count = 0

while (count < 9):

    print ('The count is:', count)

    count = count + 1

print ("Good bye!")
```

OUTPUT:

The count is: 0

The count is: 1

The count is: 2

The count is: 3

The count is: 4

The count is: 5

The count is: 6

The count is: 7

The count is: 8

Good bye!

```
count = 0

while count < 7:

    print (count, " is less than 7")

    count = count + 1

else:

    print ("Loop End")
```

OUTPUT:

0 is less than 7

1 is less than 7

2 is less than 7

3 is less than 7

4 is less than 7

5 is less than 7

6 is less than 7

Loop End

for Loop Statements:

```
for i in range(0,30, 3):
```

```
    print(i)
```

```
for i in range(0,50, 5):
```

```
    print(i)
```

OUTPUT:

0

3

6

9

12

15

18

21

24

27

0

5

10

15

20

25

30

35

40

45

Loop Control Statements:

Break-statement:

```
for i in range(1, 6):
```

```
    if i == 3:
```

```
        break
```

```
    print(i)
```

output:

1

2

continue -statement:

```
for i in range(1, 6):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```

output:

1

2

4

5

Pass statement:

```
for i in range(1,6):
```

```
    pass
```

```
print("completed loop")
```

output:

completed loop

Python format() Method:

Old String formate:

```
name = "John"
```

```
print("Hello, %s!" % name)
```

output:

Hello, John!

swapping:

```
x = int(input("Enter value of x: "))
```

```
y = int(input("Enter value of y: "))
```

```
temp = x
```

```
x = y
```

```
y = temp
```

```
print("The value of x after swapping: {}".format(x))
```

```
print("The value of y after swapping: {}".format(y))
```

output:

new string formate:

```
name = "John"
```

```
age = 23
```

```
print("My name is {} and I am {} years old.".format(name, age))
```

output:

My name is John and I am 23 years old.

Python program to do arithmetical operations:

```
num1 = int(input('Enter first number: '))
```

```
num2 = int(input('Enter second number: '))
```

```
# Add two numbers
```

```
sum = float(num1) + float(num2)
```

```
# Subtract two numbers
```

```
min = float(num1) - float(num2)
```

```
# Multiply two numbers
```

```
mul = float(num1) * float(num2)
```

```
#Divide two numbers
```

```
div = float(num1) / float(num2)
```

Python Program to Check if a Number is Odd or Even:

```
num = int(input("Enter a number: "))  
  
if (num % 2) == 0:  
  
    print("{0} is Even number".format(num))  
  
else:  
  
    print("{0} is Odd number".format(num))
```

output:

enter number 2

the number is even

Python Program to check if a Number is Positive, Negative or Zero:

```
num = float(input("Enter a number: "))  
  
if num > 0:  
  
    print("{0} is a positive number".format(num))  
  
elif num == 0:  
  
    print("{0} is zero".format(num))  
  
else:
```

```
print("{0} is negative number".format(num))
```

output:

enter a number 10,

the number is positive